

CSS Functions Reference

A Complete Guide to CSS Functions

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1 Color Functions

1.1 rgb()

Defines color using Red, Green, Blue values (0-255).

```
1 background-color: rgb(255, 0, 0);           /* Red */
2 background-color: rgb(0, 255, 0);          /* Green */
3 background-color: rgb(0, 0, 255);          /* Blue */
4 background-color: rgb(255, 255, 255);      /* White */
5 background-color: rgb(128, 128, 128);      /* Gray */
```

1.2 rgba()

RGB with alpha channel for transparency (0-1).

```
1 background-color: rgba(255, 0, 0, 0.5);    /* Red 50% */
2 background-color: rgba(0, 0, 0, 0.3);      /* Black 30% */
3 background-color: rgba(255, 255, 255, 1);  /* Opaque white */
```

1.3 hsl()

Defines color using Hue (0-360), Saturation (0-100%), Lightness (0-100%).

```
1 background-color: hsl(0, 100%, 50%);       /* Red */
2 background-color: hsl(120, 100%, 50%);     /* Green */
3 background-color: hsl(240, 100%, 50%);     /* Blue */
4 background-color: hsl(60, 100%, 50%);      /* Yellow */
5 background-color: hsl(0, 0%, 50%);         /* Gray */
```

1.4 hsla()

HSL with alpha channel for transparency.

```
1 background-color: hsla(0, 100%, 50%, 0.5); /* Red 50% */
2 background-color: hsla(120, 100%, 50%, 0.7); /* Green 70% */
3 color: hsla(240, 100%, 50%, 0.8);          /* Blue 80% */
```

1.5 hwb()

Hue, Whiteness, Blackness color model.

```
1 color: hwb(0 20% 20%);                     /* Red */
2 color: hwb(120 20% 20%);                   /* Green */
3 color: hwb(240 20% 20%);                   /* Blue */
```

2 Transform Functions

2.1 translate()

Moves element horizontally and/or vertically.

```
1 transform: translate(10px, 20px);          /* X, Y */
2 transform: translate(50%, 50%);            /* Percentage */
3 transform: translateX(100px);              /* X only */
4 transform: translateY(50px);               /* Y only */
```

2.2 scale()

Resizes element horizontally and/or vertically.

```
1 transform: scale(1.5);           /* Scale all */
2 transform: scale(1.5, 2);        /* X, Y */
3 transform: scaleX(2);            /* X only */
4 transform: scaleY(0.5);          /* Y only */
```

2.3 rotate()

Rotates element by specified angle.

```
1 transform: rotate(45deg);        /* Degrees */
2 transform: rotate(0.785rad);     /* Radians */
3 transform: rotate(0.125turn);    /* Turns */
4 transform: rotate(200grad);      /* Gradians */
```

2.4 skew()

Skews element horizontally and/or vertically.

```
1 transform: skew(20deg, 10deg);   /* X, Y */
2 transform: skewX(45deg);         /* X only */
3 transform: skewY(30deg);         /* Y only */
```

2.5 matrix()

Applies 2D transformation matrix.

```
1 transform: matrix(a, b, c, d, e, f);
2 transform: matrix(1, 0, 0, 1, 0, 0); /* Identity */
3 transform: matrix(2, 0, 0, 2, 0, 0); /* Scale 2x */
```

2.6 3D Transform Functions

```
1 transform: translate3d(10px, 20px, 30px); /* X, Y, Z */
2 transform: translateZ(50px);              /* Z only */
3 transform: scale3d(1.5, 1.5, 2);          /* 3D scale */
4 transform: scaleZ(1.5);                   /* Z only */
5 transform: rotate3d(1, 1, 1, 45deg);      /* Axis, angle */
6 transform: rotateX(45deg);                /* X rotation */
7 transform: rotateY(45deg);                /* Y rotation */
8 transform: rotateZ(45deg);                /* Z rotation */
9 transform: matrix3d(a, b, c, d, e, f,
10                  g, h, i, j, k, l,
11                  m, n, o, p);             /* 3D matrix */
```

3 Filter Functions

3.1 blur()

Applies Gaussian blur effect.

```
1 filter: blur(5px);
2 filter: blur(0px);           /* No blur */
3 filter: blur(10px);          /* Heavy blur */
```

3.2 brightness()

Adjusts brightness (0-2+, 1=normal).

```
1 filter: brightness(0.5);     /* 50% brightness */
2 filter: brightness(1);       /* Normal */
3 filter: brightness(1.5);     /* 150% brightness */
```

3.3 contrast()

Adjusts contrast (0-2+, 1=normal).

```
1 filter: contrast(0.5);       /* Low contrast */
2 filter: contrast(1);         /* Normal */
3 filter: contrast(2);         /* High contrast */
```

3.4 grayscale()

Converts to grayscale (0-1, 0=color).

```
1 filter: grayscale(0);        /* Full color */
2 filter: grayscale(0.5);      /* 50% grayscale */
3 filter: grayscale(1);        /* Full grayscale */
```

3.5 hue-rotate()

Rotates hue on color wheel.

```
1 filter: hue-rotate(0deg);     /* No rotation */
2 filter: hue-rotate(90deg);    /* 90° rotation */
3 filter: hue-rotate(180deg);   /* Invert colors */
```

3.6 invert()

Inverts colors (0-1).

```
1 filter: invert(0);           /* Normal */
2 filter: invert(0.5);         /* 50% inverted */
3 filter: invert(1);           /* Fully inverted */
```

3.7 opacity()

Adjusts opacity (0-1).

```
1 filter: opacity(0.5);        /* 50% transparent */
2 filter: opacity(0);          /* Invisible */
3 filter: opacity(1);          /* Opaque */
```

3.8 saturate()

Adjusts saturation (0-2+, 1=normal).

```
1 filter: saturate(0);           /* Grayscale */
2 filter: saturate(0.5);        /* Desaturated */
3 filter: saturate(1);          /* Normal */
4 filter: saturate(2);          /* Oversaturated */
```

3.9 sepia()

Applies sepia tone (0-1).

```
1 filter: sepia(0);             /* No sepia */
2 filter: sepia(0.5);          /* 50% sepia */
3 filter: sepia(1);            /* Full sepia */
```

3.10 drop-shadow()

Applies drop shadow effect.

```
1 filter: drop-shadow(5px 5px 10px rgba(0,0,0,0.5));
2 filter: drop-shadow(2px 2px 4px black);
3 filter: drop-shadow(0 0 20px rgba(0,0,0,0.8));
```

4 Background Functions

4.1 linear-gradient()

Creates linear gradient backgrounds.

```
1 background: linear-gradient(to right, red, blue);
2 background: linear-gradient(45deg, red, green, blue);
3 background: linear-gradient(to bottom,
4                               rgba(0,0,0,0),
5                               rgba(0,0,0,1));
6 background: linear-gradient(90deg,
7                               red 0%,
8                               yellow 50%,
9                               blue 100%);
```

4.2 radial-gradient()

Creates radial gradient backgrounds.

```
1 background: radial-gradient(red, blue);
2 background: radial-gradient(circle, red, blue);
3 background: radial-gradient(ellipse at top left,
4                               red, blue);
5 background: radial-gradient(circle at 50% 50%,
6                               red 0%,
7                               blue 100%);
```

4.3 conic-gradient()

Creates conic (circular) gradient.

```
1 background: conic-gradient(red, yellow, green,  
2                           cyan, blue, magenta, red);  
3 background: conic-gradient(from 45deg, red, blue);  
4 background: conic-gradient(at 50% 50%,  
5                           red 0deg,  
6                           blue 90deg);
```

4.4 repeating-linear-gradient()

Repeats linear gradient pattern.

```
1 background: repeating-linear-gradient(45deg,  
2                                     red,  
3                                     blue 20px);  
4 background: repeating-linear-gradient(to right,  
5                                     #ff0000 0px,  
6                                     #ffff00 10px);
```

4.5 repeating-radial-gradient()

Repeats radial gradient pattern.

```
1 background: repeating-radial-gradient(red 0px,  
2                                     blue 10px);  
3 background: repeating-radial-gradient(circle at  
4                                     center,  
5                                     rgba(255,0,0,0.8) 0px,  
6                                     rgba(255,0,0,0) 20px);
```

4.6 url()

References external images/resources.

```
1 background-image: url('image.png');  
2 background-image: url('../images/bg.jpg');  
3 background-image: url('data:image/png;base64,...');
```

5 Shape Functions

5.1 circle()

Defines circular shape for clip-path.

```
1 clip-path: circle(50%);           /* Full circle */
2 clip-path: circle(50px);          /* Fixed size */
3 clip-path: circle(50% at 50% 50%); /* Center */
4 clip-path: circle(100px at 10% 20%); /* Custom position */
```

5.2 ellipse()

Defines elliptical shape for clip-path.

```
1 clip-path: ellipse(50% 30%);
2 clip-path: ellipse(100px 50px);
3 clip-path: ellipse(50% 30% at 50% 50%);
```

5.3 polygon()

Defines polygonal shape for clip-path.

```
1 clip-path: polygon(50% 0%, 100% 50%,
2                 50% 100%, 0% 50%); /* Diamond */
3 clip-path: polygon(0% 0%, 100% 0%,
4                 100% 100%, 0% 100%); /* Rectangle */
5 clip-path: polygon(50% 0%, 75% 50%,
6                 50% 100%, 25% 50%); /* Diamond */
```

5.4 inset()

Defines inset rectangular shape.

```
1 clip-path: inset(0);           /* No inset */
2 clip-path: inset(10px);        /* Inset all sides */
3 clip-path: inset(10px 20px);   /* Top/bottom, left/right */
4 clip-path: inset(10px 20px 30px 40px); /* Individual sides */
5 clip-path: inset(10px round 5px); /* With rounded corners */
```

6 Timing Functions

6.1 cubic-bezier()

Defines custom timing function for animations.

```
1 animation-timing-function: cubic-bezier(0.25, 0.1,
2                                     0.25, 1);
3 /* Common presets */
4 cubic-bezier(0.25, 0.1, 0.25, 1); /* ease */
5 cubic-bezier(0, 0, 1, 1);          /* linear */
6 cubic-bezier(0.42, 0, 1, 1);       /* ease-in */
7 cubic-bezier(0, 0, 0.58, 1);       /* ease-out */
8 cubic-bezier(0.42, 0, 0.58, 1);    /* ease-in-out */
```

6.2 steps()

Divides animation into steps.

```
1 animation-timing-function: steps(5);  
2 animation-timing-function: steps(10, start);  
3 animation-timing-function: steps(4, end);
```

7 Math Functions

7.1 calc()

Performs mathematical calculations for values.

```
1 width: calc(100% - 20px);  
2 height: calc(100vh - 50px);  
3 font-size: calc(16px + 2vw);  
4 margin: calc(10px * 2);  
5 padding: calc(1em + 5px);  
6 flex-basis: calc(100% / 3 - 20px);
```

7.2 min()

Returns minimum value from list.

```
1 width: min(100%, 500px);  
2 font-size: min(5vw, 2em);  
3 padding: min(10vw, 50px);
```

7.3 max()

Returns maximum value from list.

```
1 width: max(100px, 50%);  
2 font-size: max(1rem, 2vw);  
3 padding: max(1em, 20px);
```

7.4 clamp()

Constrains value between minimum and maximum.

```
1 width: clamp(200px, 50%, 800px);  
2 font-size: clamp(1rem, 5vw, 3rem);  
3 padding: clamp(10px, 2vw, 50px);
```

8 Animation Functions

8.1 var()

References custom CSS properties (variables).

```
1 --main-color: #06c;
2 --padding: 10px;
3 color: var(--main-color);
4 padding: var(--padding);
5 margin: var(--padding) var(--padding) 0 0;
6 background: var(--main-color, blue); /* Fallback */
```

8.2 attr()

Returns attribute value of element.

```
1 content: attr(data-tooltip);
2 content: "(" attr(href) ";";
3 width: attr(data-width);
```

9 Typography Functions

9.1 format()

Specifies font format in @font-face.

```
1 @font-face {
2     font-family: "MyFont";
3     src: url("font.woff2") format("woff2"),
4         url("font.woff") format("woff");
5 }
```

10 Filter Function Combinations

Combine multiple filter functions:

```
1 filter: blur(5px) brightness(1.2) contrast(1.5);
2 filter: grayscale(0.5) sepia(0.3);
3 filter: hue-rotate(90deg) saturate(1.5)
4         opacity(0.8);
5 filter: brightness(0.8) contrast(1.2)
6         drop-shadow(5px 5px 10px rgba(0,0,0,0.3));
```

11 Transform Function Combinations

Chain multiple transforms:

```
1 transform: translateX(50px) scale(1.5)
2           rotate(45deg);
3 transform: translate(10px, 20px)
4           rotateZ(45deg)
5           scale(1.2);
6 transform: perspective(1000px) rotateX(45deg)
7           rotateY(45deg);
```

12 Browser Support Summary

```
1 /* Universal Support (All modern browsers) */
2 rgb(), rgba(), hsl(), hsla()
3 translate(), scale(), rotate(), skew()
4 linear-gradient(), radial-gradient()
5 calc(), var(), attr()
6
7 /* Wide Support (Chrome, Firefox, Safari, Edge 15+) */
8 3D transforms, filters, cubic-bezier()
9 conic-gradient(), min(), max(), clamp()
10
11 /* Newer (Check caniuse.com) */
12 hwb(), repeating-conic-gradient()
```

13 Quick Reference Table

```
1 /* Color Functions */
2 rgb(r, g, b) / rgba(r, g, b, a)
3 hsl(h, s%, l%) / hsla(h, s%, l%, a)
4 hwb(h w% b%)
5
6 /* Transform Functions */
7 translate(x, y) / translate3d(x, y, z)
8 scale(s) / scaleX(sx) / scaleY(sy) / scaleZ(sz)
9 rotate(angle) / rotateX/Y/Z(angle)
10 skew(x, y) / skewX(x) / skewY(y)
11
12 /* Filter Functions */
13 blur() | brightness() | contrast() | grayscale()
14 hue-rotate() | invert() | opacity() | saturate()
15 sepia() | drop-shadow()
16
17 /* Gradient Functions */
18 linear-gradient() / radial-gradient()
19 conic-gradient() / repeating-*-gradient()
20
21 /* Shape Functions */
22 circle() | ellipse() | polygon() | inset()
23
24 /* Math Functions */
25 calc() | min() | max() | clamp()
26
27 /* Timing Functions */
28 cubic-bezier() | steps()
29
30 /* Variable Functions */
31 var(--custom-property) | attr(attribute)
```